Fabrication and Integration Process

1. Material Choice Justification

Every material used in Claw & Order was deliberately chosen to support the machine's arcade aesthetic, structural integrity, and ease of fabrication. Below is a full breakdown of each key material decision.

1.1 Acrylic vs. MDF — Structural Panels & Base

Category	Chosen Material	Why Not the Alternative?
Aesthetics	Acrylic — clean, glossy, arcade-like finish	MDF — dull, matte, plain surface
Weight	Acrylic — lighter for the base panel	MDF — heavier, adds unnecessary mass
Finish	Acrylic — smooth out of the box, no sanding	MDF — requires priming and painting to look clean
Strength	Acrylic — strong and rigid for flat panels	MDF — prone to swelling and crumbling at edges
Moisture	Acrylic — unaffected by humidity	MDF — absorbs moisture and warps over time

Acrylic was selected for the base of the claw machine due to its superior visual finish, lighter weight, and durability. Its transparent or coloured variants reinforced the arcade-style aesthetic the team was targeting. MDF, while cheaper, would have required additional finishing steps and is less durable in the long term.

1.2 PLA vs. Resin — 3D Printed Components

Category	Chosen Material	Why Not the Alternative?
Ease of Print	PLA — beginner-friendly, low warp	Resin — requires UV curing and post-processing
Strength	PLA — strong enough for pulleys and mounts	Resin — brittle under mechanical stress
Turnaround	PLA — fast print-to-use pipeline	Resin — longer process with washing and curing
Safety	PLA — safe, low odour	Resin — toxic fumes, requires PPE
Cost	PLA — inexpensive and widely available	Resin — significantly more expensive per unit

PLA was chosen for all 3D printed components including the CoreXY pulley system, button housings, structural supports, and the claw itself. It offered the right balance of printability and mechanical strength. The claw was printed from a template and required no post-processing, making PLA the clear practical choice over resin for a project of this scale.

1.3 Laser-Cut MDF vs. Acrylic — Exterior Frame

The exterior frame panels were laser cut from MDF rather than acrylic. For the structural frame, MDF was preferred as it is easier to nail and glue, holds screws well, and provides a rigid skeleton. The laser cutter produced clean, precise edges which were then assembled using wood glue and nails. The frame was later painted in the machine's signature red and blue arcade colour scheme, which effectively masked MDF's raw appearance.

2. Fabrication Steps — Build Log

The following build log documents the complete fabrication and integration process for Claw & Order in chronological order, from raw materials to a fully operational machine.

Step	Stage	Details

01	Frame Design & Laser Cutting	Designed exterior frame panels and laser cut them from MDF sheets. Precision cuts ensured clean joints and consistent dimensions across all panels.
02	Frame Assembly	Assembled all laser cut panels using wood glue and nails. Panels were clamped and left to set, forming the full exterior box structure of the claw machine.
03	3D Printing — Pulley System	Printed CoreXY pulley system components in PLA including pulley mounts, belt guides, and corner brackets. Multiple iterations were printed to account for dimensional tolerances.
04	3D Printing — Claw	Printed the claw mechanism from a template file in PLA. Tested mechanical movement of claw fingers before integration.
05	3D Printing — Supports & Buttons	Printed additional structural supports, button housings, and mounting brackets required for electronics integration.
06	Acrylic Base Cutting	Cut the acrylic sheet to size for the machine's base panel. Chosen for its aesthetic finish and light weight.
07	CoreXY Gantry Assembly	Assembled the CoreXY gantry system using 3D printed pulleys and rods. Initial challenges with rod dimensions required adjustments to printed mounts to achieve smooth movement.
08	Belt Routing — Iteration 1	Attempted to route kite flying nylon thread through the pulley system. Thread proved too stiff and difficult to loop correctly through the CoreXY path.
09	Belt Routing — Iteration 2	Replaced kite thread with a nylon clothes line. While more flexible, the thickness and texture made it difficult to handle and tension correctly within the pulleys.
10	Motor Mounting — X & Y Axes	Mounted two DC motors for the X and Y axes of the CoreXY gantry. Motors secured using 3D printed mounts and connected to the belt system.
11	Motor Mounting — Z Axis	Mounted the third DC motor for the Z axis to control up and down movement of the claw assembly.
12	Motor Mounting — Claw	Mounted the fourth DC motor to control opening and closing of the claw. Originally planned as a servo motor; switched to DC motor after servo testing revealed mechanical failure.
13	Individual Component Testing	Tested each electronic component individually: pushbuttons, DC motors, servo (later replaced), NeoPixel LED strip, and motor drivers. Verified GPIO pin assignments on the ESP32.

14	Motor Driver Wiring	Wired two L298N motor drivers to control all four DC motors. Each driver handles two motors. ENA/ENB jumpers removed to enable PWM speed control.
15	Input Controls	Initially wired a joystick for XY navigation. After joystick malfunctions, switched to six pushbuttons: four for directional movement, one for Z axis, one for claw open/close.
16	NeoPixel Integration	Wired 16-LED WS2812 NeoPixel strip to GPIO 12. Programmed colour states: green for idle, blue for movement, yellow for Z axis, red for claw closing.
17	Soldering & Wire Management	Soldered key connections to improve reliability. Encountered issues with internal wire breakage, disconnecting wires, and incorrect solder joints which required rework.
18	Software — Component Testing	Wrote and tested individual MicroPython scripts for each motor, buttons, and NeoPixel before integrating into a unified codebase.
19	Software — CoreXY Logic	Implemented CoreXY kinematics: left/right moves both motors in the same direction; up/down moves them in opposite directions.
20	Software — Claw Timing	Calibrated DC motor timing for the claw open/close cycle. Adjusted time.sleep() values to ensure correct angle without straining the motor.
21	Final Assembly	Integrated all mechanical and electronic components into the completed frame. Secured all wiring, mounted the control panel, and performed full system tests.
22	Painting & Finishing	Painted the exterior frame in the machine's signature bright red and blue arcade colour scheme.
23	Final Testing	Conducted end-to-end testing of all functions: XY movement, Z axis, claw open/close, NeoPixel feedback. Verified all systems operated correctly together.

3. Key Challenges & How They Were Resolved

3.1 Mechanical Challenges

- CoreXY rod dimensions required multiple iterations of 3D printed mounts before achieving smooth gantry movement.
- Kite nylon thread was too stiff to route through the pulley system and was replaced with a nylon clothes line, which also presented handling challenges due to its thickness and texture.

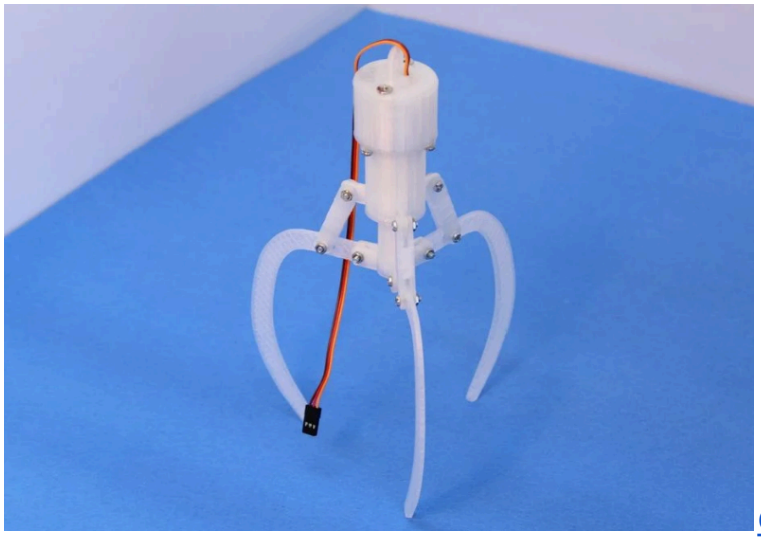
3.2 Electrical Challenges

- Internal wire breakage discovered mid-testing, requiring continuity tests on all jumper wires.
- Disconnecting wires at breadboard and L298N terminals caused intermittent motor failures that were difficult to diagnose.
- Incorrect soldering at several joints required rework to establish reliable connections.

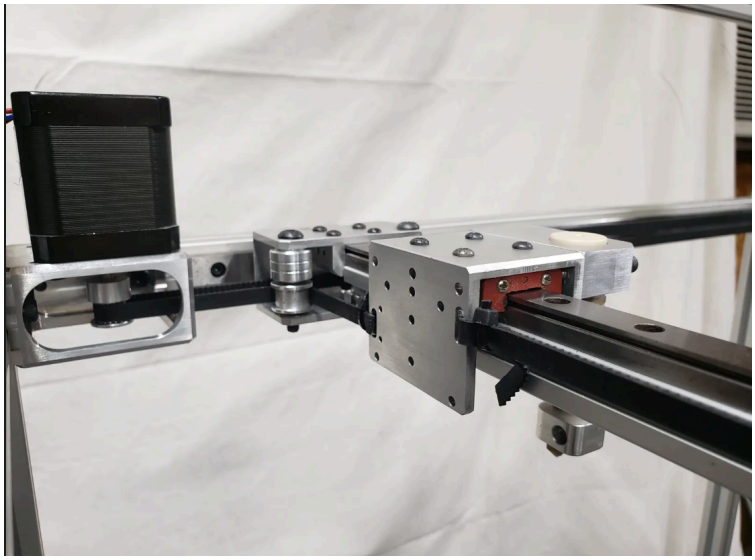
3.3 Design Iteration Challenges

- The original joystick malfunctioned and was replaced with a 4-button directional control system.
- The servo motor planned for the claw was found to be mechanically seized and was replaced with a DC motor, requiring code and wiring changes.
- Claw motor timing required careful calibration of sleep intervals to achieve correct open and close angles without damaging the mechanism.

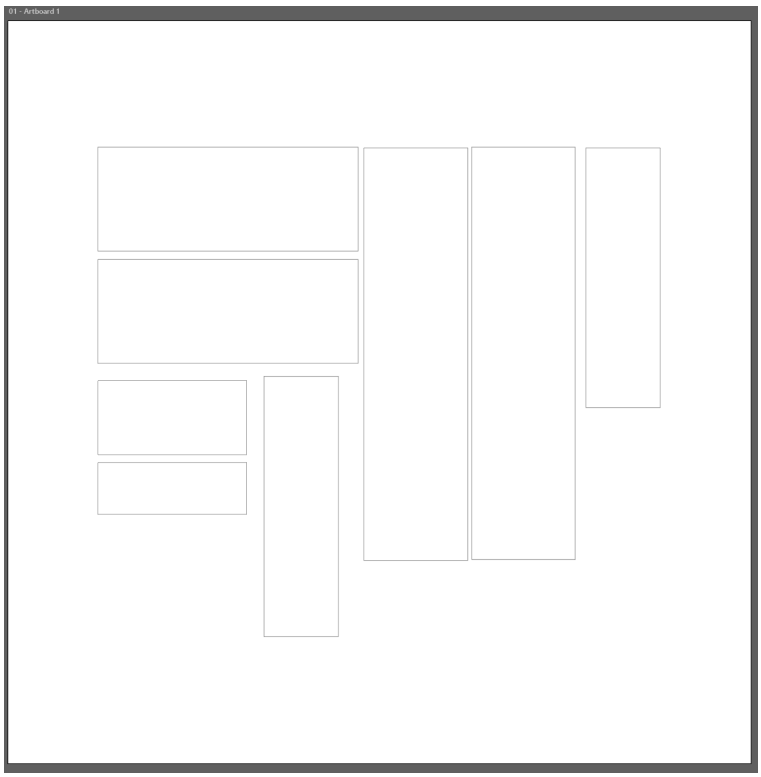
4. Models and Files



Claw 3D Model

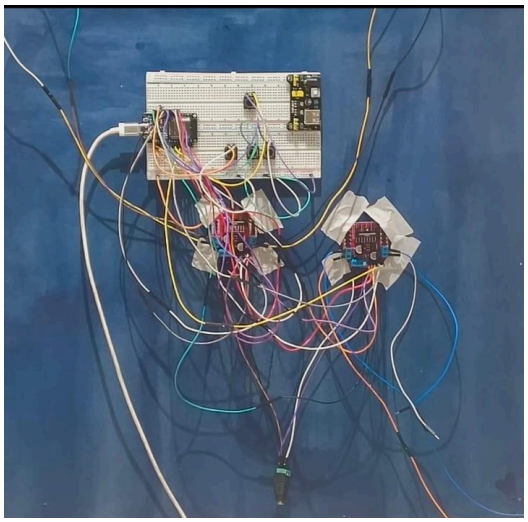


CoreXY Carriage Reference



A laser cutting file for the exterior

In progress Claw Machine



In progress circuit